A Leakage-Aware Benchmark for Ultra-Small Language Models on Structured Speech-Command Routing

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Abstract. *Deploying structured command routing locally on mobile devices requires ultra-small language models ($\leq$ 3B parameters). We present a reproducible benchmark evaluating five sub-3B instruction-tuned models across distinct families on a transcript-to-JSON routing task derived from Fluent Speech Commands (FSC). We evaluate under an explicit two-tool contract with balanced policy abstention (50% calls, 50% abstentions), comparing the official speaker-disjoint split against an audited phrase-disjoint partition (n = 2,296, 38 template clusters). Under the strict contract, a transparent lexical baseline achieves 77.40% exact match, outperforming every zero-shot neural model. Qwen3.5-2B attains 100% validity but collapses to universal abstention ($R_{call}^{exact} = 0\%$), while three models score 0% contract validity due to unconstrained schema deviations. We show this structural failure is resolved by supervised adaptation (LoRA), establishing crucial guidelines for edge SLM evaluation.*

1. Introduction and Scope

Deploying natural language command routing locally on edge hardware requires ultra-small language models (SLMs, $\leq$ 3B parameters). However, evaluating these models zero-shot introduces two severe pitfalls: (i) template leakage across speakers in benchmark splits, and (ii) degenerate policy compliance, where models achieve seemingly high accuracy by trivial abstention.

We evaluate five checkpoints within the sub-3B on-device regime [4]: Qwen2.5-0.5B, Qwen3.5-0.8B/2B [7], TinyLlama-1.1B [5], and SmolLM2-1.7B [6]. The input is a transcription from Fluent Speech Commands (FSC) [1], and the output is a validated JSON object conforming to a strict two-tool contract: `media_control` and `volume_adjust`, with all other 27 native intents mapped to a derived abstention target. The evaluation is balanced 1:1 (7,478 calls and 7,478 abstentions per split). We evaluate on both FSC's official speaker-disjoint split (n = 1,934) and an audited phrase-disjoint split (n = 2,296, 38 template clusters) where no transcription template crosses partitions.

2. Experimental Results

As shown in Table 1 and Figure 1, the lexical control dominates all zero-shot models. Exact Call Recall (R_{call}^{exact}) denotes the ratio of correct calls with exact tool and arguments over gold call instances. Strict Template Exact denotes the proportion of template clusters where 100% of items are correctly routed. Qwen3.5-2B attains 100% validity and 50.00% exact match, but inspection reveals complete policy collapse: it outputs `abstain` on 100% of inputs, yielding $R_{call}^{exact} = 0\%$. Qwen3.5-0.8B is the only model attempting active

Table 1. Phrase-disjoint vs. Official test results (%).

	Phrase-Disjoint (Unseen Templates)			Official (Speaker Split)		
System	**Valid**	**Exact**	**Exact Call Recall**	**Valid**	**Exact**	**Template Exact**
Always abstain	100.00	50.00	0.00	100.00	50.00	75.71
Lexical control	100.00	**77.40**	**54.79**	100.00	**77.92**	**89.07**
Qwen2.5-0.5B	0.00	0.00	0.00	0.00	0.00	0.00
Qwen3.5-0.8B	60.15	43.60	33.45	73.47	54.65	56.28
TinyLlama-1.1B	0.00	0.00	0.00	0.00	0.00	0.00
SmolLM2-1.7B	0.00	0.00	0.00	0.78	0.00	0.00
Qwen3.5-2B	100.00	50.00	0.00	100.00	50.98	76.11

routing, but exhibits an observed split gap of 11.05 pp from official (54.65%) to phrase-disjoint (43.60%). Because the phrase-disjoint cluster bootstrap 95% CI is wide ([26.02%, 63.35%], over 38 clusters with 9 call and 29 abstain clusters), this gap is descriptive rather than a formal causal test.

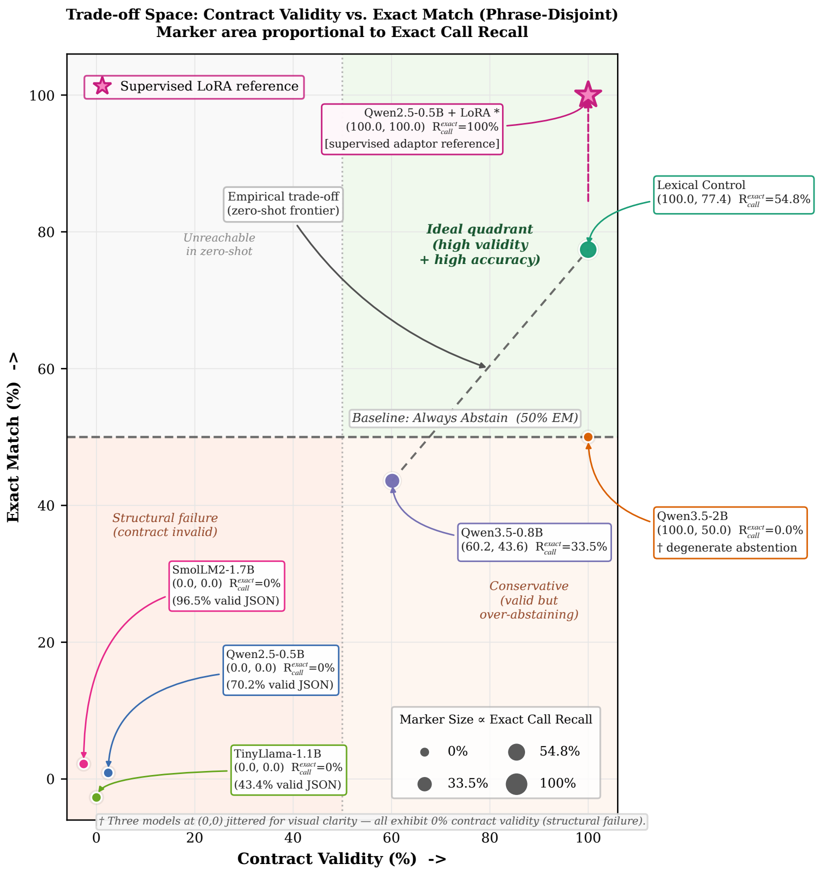
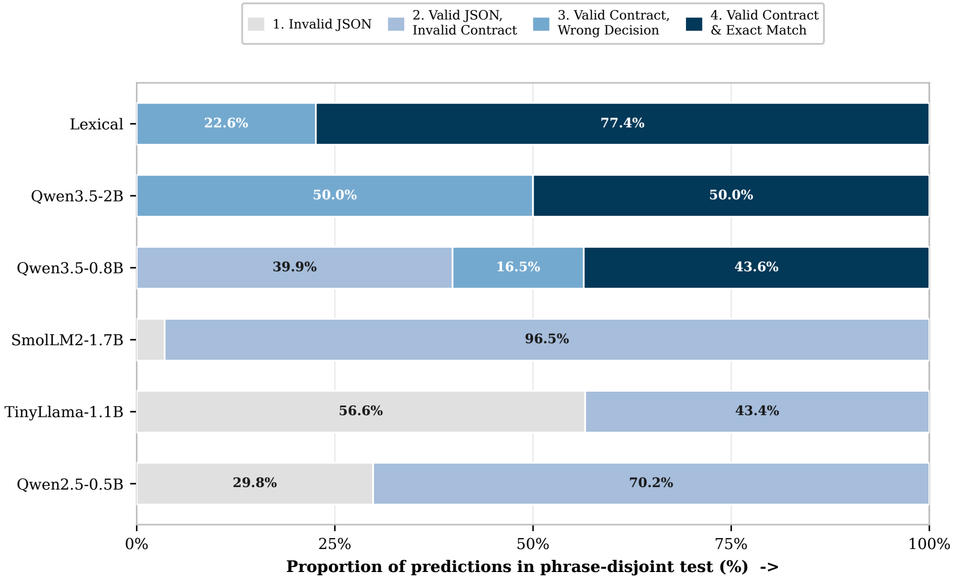


Figure 1. Trade-off space on phrase-disjoint test. Marker area ∝ exact call recall. The lexical baseline dominates zero-shot SLMs; Qwen3.5-2B collapses to 100% abstentions.

Figure 2 reveals the exact point of structural failure: SmolLM2-1.7B produces 96.47% syntactically parseable JSON, but 0% valid contract objects due to schema deviations (missing action/tool keys or invalid parameters). Unconstrained zero-shot decoding is insufficient for sub-3B models to adhere to rigid task schemas.



Stages: 1 Invalid JSON -> 2 Valid JSON but invalid contract -> 3 Valid contract but wrong decision -> 4 Valid contract and exact match. Values shown for slices >= 8%.

Figure 2. Structural cascade decomposing outputs into: (1) invalid JSON, (2) valid JSON but invalid schema, (3) valid contract but wrong call, and (4) exact match.

3. Discussion, Adaptation, and Conclusions

Two orthogonal techniques address this bottleneck: (i) **Grammar-Constrained Decoding** (e.g., GBNF in `llama.cpp`, Outlines [9]), which forces syntactic compliance by construction (collapsing stages 1–2 of Figure 2 to zero, though semantic discrimination still relies on the model); and (ii) **Supervised LoRA Adaptation** [8]: an in-distribution Qwen2.5-0.5B LoRA adaptor reaches 100% contract validity and 100% exact match, demonstrating that the contract is readily learnable with parameter-efficient fine-tuning.

In conclusion, zero-shot structured routing without grammar guidance is fragile in the ultra-small regime. Benchmarks must decouple syntactic validity from semantic utility, audit template leakage, and incorporate transparent non-neural baselines. Extending this evaluation to multi-domain SLU benchmarks like SLURP [2] and compositional parsing corpora like STOP [3] represents an essential next step.

References

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